[Paper 145 — A Memoir upon Caustics, continued]

Printed pages 360–371; scan PDF pages 376–387.

Now

$$\frac{dP}{dx} = 2\{(4a^2-1)x - a\}, \qquad \frac{dP}{dy} = 2(4a^2-1)y,$$

$$\frac{dQ}{dx} = 2axy, \qquad \frac{dQ}{dy} = a(x^2+3y^2-a^2) = a(S+2y^2);$$

substituting these values in the last preceding equation, we find

$$\frac{(4a^2-1)x-a}{(4a^2-1)y} = \frac{2xy}{S+2y^2},$$

or, reducing,

$$(4a^2-1)x - a = \frac{2axy^2}{S};$$

and using this to simplify the equation

$$P^2\frac{dP}{dx} - 18Q\,\frac{dQ}{dx} = 0,$$

we have

$$P^2\frac{4axy^2}{S} - 18axyS\cdot 2axy = 0,$$

i.e.

$$\frac{P^2}{S} - 9axS = 0,$$

and therefore

$$9x = \frac{P^2}{aS^2}.$$

Multiplying by P and writing for P^3 its value $27a^3y^2S^3$, we have

$$Px = 3ayS,$$

and thence

$$P^2 = \frac{9a^2y^2S^2}{x^2}, \qquad P^3 = \frac{27a^3y^3S^3}{x^3} = 27a^3y^2S^3,$$

whence

$$\frac{ay^2}{x^3} = S^2, \quad \frac{a^{\frac{1}{3}}y^{\frac{2}{3}}}{x} = S, \quad \text{or } \frac{y^2}{S^3} = \frac{x^3}{a^3};$$

and substituting in the equation

$$x = \frac{a}{4a^2-1}\left(1+\frac{2y^2}{S}\right),$$

we find

$$x = \frac{a}{4a^2-1}\left(1+\frac{2x}{a^{\frac{1}{3}}}\right),$$

or, rationalising,

$$4ax^3 - \{(4a^2-1)x-a\}^3 = 0,$$

or, what is the same thing,

$$(4ax-1)\big(x-a(a+i\sqrt{1-a^2})^3\big)\big(x-a(a-i\sqrt{1-a^2})^3\big)=0.$$

The factor $4ax-1$ equated to zero gives $x=\frac{1}{4a}$, from which y may be found, but the resulting point is not a double point; the other factors give each of them double points, and if we write

$$x=a(a+i\sqrt{1-a^2})^3,$$

we find

$$y=\frac{2a^2i(a+i\sqrt{1-a^2})^{\frac{5}{2}}}{(3a-i\sqrt{1-a^2})^{\frac{1}{2}}},$$

values which, in fact, belong to one of the four double points. It is easy to see that the points in question are always imaginary.

It may be noticed, by way of verification, that the preceding values of x, y give

$$\begin{aligned}(4a^2-1)(x^2+y^2)-2ax-a^2&=\frac{a^2}{1+8a^2}\big(1-4a^2-4ai\sqrt{1-a^2}\big),\\ x^2+y^2-a^2&=\frac{4a^2}{1+8a^2}(3a-i\sqrt{1-a^2})(a+i\sqrt{1-a^2})^3,\\ y^2&=\frac{4a^4}{1+8a^2}\big(-1+14a^2-16a^4+2a(3-8a^2)i\sqrt{1-a^2}\big);\end{aligned}$$

and if the quantities within ( ) on the right-hand side are represented by A, B, C, then

$$\frac{A}{B}=-(a+i\sqrt{1-a^2}),$$

$$\frac{C}{B}=-(a+i\sqrt{1-a^2})^3,$$

whence we have identically,

$$\left(\frac{A}{B}\right)^3=\frac{C}{B},\ \text{ or } A^3=B^2C,$$

by means of which it appears that the values of x, y satisfy, as they should do, the equation of the caustic; and by forming the expressions for $(4a^2-1)x-a$ and $x^2+2y^2-a^2$, it might be shown, *à posteriori*, that the point in question was a double point.

XXIII.

The equation

$$\{(4a^2-1)(x^2+y^2)-2ax-a^2\}^3-27a^2y^2(x^2+y^2-a^2)^2=0$$

becomes when $a=1$ (i.e. when the radiant point is in the circumference),

$$\{3y^2+(x-1)(3x+1)\}^3-27y^2(y^2+x^2-1)^2=0;$$

it is easy to see that this divides by $(x-1)^2$; and throwing out this factor, we have for the caustic the equation of the fourth order,

$$27y^4+18y^2(3x^2-1)+(x-1)(3x+1)^3=0.$$

XXIV.

The equation

$$\{(4a^2-1)(x^2+y^2)-2ax-a^2\}^3-27a^2y^2(x^2+y^2-a^2)^2=0$$

becomes when $a=\infty$ (i.e. in the case of parallel rays),

$$(4x^2+4y^2-1)^3-27y^2=0,$$

which may also be written

$$64x^6+48x^4(4y^2-1)+12x^2(4y^2-1)^2+(8y^2+1)^2(4y^2-1)=0.$$

XXV.

It is now easy to trace the curve. Beginning with the case $a=\infty$, the curve lies wholly within the reflecting circle, which it touches at two points; the line joining the points of contact, being in fact the axis of y, divides the curve into two equal portions; the curve has in the present, as in every other case (except one limiting case), two cusps on the axis of x (see fig. 6). Next, if a be positive and >1, the general form of the curve is the same as before, only the line joining the points of contact with the reflecting circle divides the curve into unequal portions, that in the

Fig. 6. $a=\infty$.

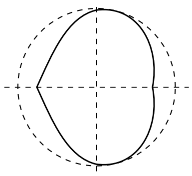
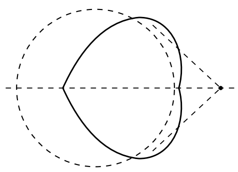


Fig. 7. $a>1$.



neighbourhood of the radiant point being the smaller of the two portions (see fig. 7). When $a=1$, the two points of contact with the reflecting circle unite together at the radiant point; the curve throws off, as it were, the two coincident lines $x=1$, and the order is reduced from 6 to 4. The curve has the form fig. 8, with only a single cusp on the axis of x. If a be further diminished, $a<1>\dfrac{1}{\sqrt{2}}$, the curve takes the form shown by fig. 9, with two infinite branches, one of them having simply a cusp on the axis of x, the other having a cusp on the axis of x, and a pair of cusps at its intersection with the circle through the radiant point; there are two asymptotes equally inclined to the axis of x. In the case $a=\dfrac{1}{\sqrt{2}}$, the form of the curve is nearly the same as before, only the cusps upon the circle through the radiant point lie on the axis of y (see fig. 10). The case $a<\dfrac{1}{\sqrt{2}}>\frac{1}{2}$ is shown, fig. 11. For $a=\frac{1}{2}$, the two asymptotes coincide with the axis of x; one of the branches of the curve has wholly disappeared, and the form of the other is modified by the coincidence of the asymptotes

Fig. 8. $a = 1$.

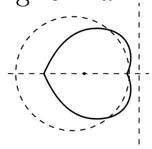


Fig. 9. $a < 1 > \frac{1}{\sqrt{2}}$.

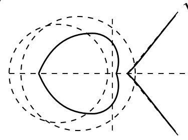


Fig. 13. $a = \frac{1}{3}$.

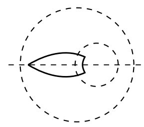


Fig. 11. $a < \frac{1}{\sqrt{2}} > \frac{1}{2}$.

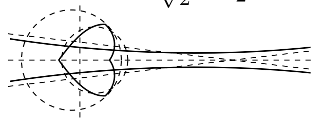


Fig. 10. $a = \frac{1}{\sqrt{2}}$.

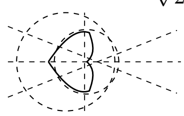
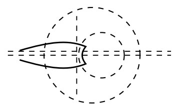


Fig. 12. $a = \frac{1}{2}$.



with the axis of x; it has in fact acquired a cusp at infinity on the axis of x (see fig. 12). When $a < \frac{1}{2}$, the curve consists of a single finite branch, with two cusps on the axis of x, and two cusps at the points of intersection with the circle through the radiant point; one of the last-mentioned cusps will be outside the reflecting circle as long as $a > \frac{1}{4}$; fig. 13 represents the case $a = \frac{1}{4}$, for which this cusp is upon the reflecting circle. For $a < \frac{1}{4}$, the curve lies wholly within the reflecting circle, one of the cusps upon the axis of x being always within, and the other always without the circle through the radiant point, and as a approaches 0 the curve becomes smaller and smaller, and ultimately disappears in a point. The case a negative is obviously included in the preceding one.

Several of the preceding results relating to the caustic by reflexion of a circle were obtained, and the curve is traced in a memoir by the Rev. Hamnet Holditch, *Quarterly Mathematical Journal*, t. I. [1857, pp. 93–111].

XXVI.

Suppose next that rays proceeding from a point are *refracted* at a circle. Take the centre of the circle as origin, let the radius be c, and take ξ, η as the coordinates of the radiant point, α, β the coordinates of the point of incidence, x, y the coordinates of a point in the refracted ray: then the general equation

$$-\overline{q}\,\overline{G}^2\,\overline{\nabla QGN} + \mu^2\overline{QG}^2\,\overline{\nabla qGN}^2 = 0$$

becomes, taking the centre of the circle as the point N on the normal, or writing $a = b$, $b = 0$,

$$-[(x-\alpha)^2 + (y-\beta)^2]\{(\alpha^2+\beta^2)\} + \mu^2(\alpha^2+\beta^2)\{(\xi-\alpha)^2 + (\eta-\beta)^2\} = 0;$$

or putting $\alpha^2 + \beta^2 = c^2$, and expanding,

$$\begin{aligned}
&\alpha^2\,[2(\eta^2 x - \mu^2 x y^2 \xi)] \\
&+ \alpha^2\beta\,[-4(\xi\eta x - \mu^2 xy\xi) + 2(\eta^2 y - \mu^2 y^2 \eta)] \\
&+ \alpha\beta^2\,[-4(\xi y - \mu^2 xy\eta) + 2(\xi^2 x - \mu^2 x^2 \xi)] \\
&+ \beta^3\,[2(\xi^2 y - \mu^2 y^2 \eta)] \\
&- \alpha^2\,[(x^2+y^2+c^2)\eta^2 - \mu^2(\xi^2+\eta^2+c^2)y^2] \\
&+ 2\alpha\beta\,\{(x^2+y^2+c^2)\xi\eta - \mu^2(\xi^2+\eta^2+c^2)xy\} \\
&- \beta^2\,\{(x^2+y^2+c^2)\xi^2 - \mu^2(\xi^2+\eta^2+c^2)x^2\} \\
&= 0,
\end{aligned}$$

which may be represented by

$$A\alpha^3 + B\alpha^2\beta + C\alpha\beta^2 + D\beta^3 + F\alpha^2 + G\alpha\beta + H\beta^2 = 0.$$

Now $\alpha^2 + \beta^2 = c^2$, and we may write

$$\alpha = \tfrac{1}{2}c\left(z + \frac{1}{z}\right), \quad \beta = \tfrac{1}{2}ci\left(z - \frac{1}{z}\right).$$

The equation thus becomes

$$A\left(z+\frac{1}{z}\right)^3 - Bi\left(z+\frac{1}{z}\right)^2\left(z-\frac{1}{z}\right) - C\left(z+\frac{1}{z}\right)\left(z-\frac{1}{z}\right)^2 - Di\left(z-\frac{1}{z}\right)^3$$
$$+\frac{2}{c}F\left(z+\frac{1}{z}\right)^2 - \frac{2}{c}Gi\left(z+\frac{1}{z}\right)\left(z-\frac{1}{z}\right) - \frac{2}{c}H\left(z-\frac{1}{z}\right)^2 = 0$$

or expanding,

$$\begin{aligned}
&(A - Bi - C - Di)\,z^3 \\
&+ \tfrac{2}{c}(F - Gi - H)\,z^2 \\
&+ (3A - Bi + C + 3Di)\,z \\
&+ \tfrac{2}{c}(F + H) \\
&+ (3A + Bi + C - 3Di)\,\tfrac{1}{z} \\
&+ \tfrac{2}{c}(F + Gi - H)\,\tfrac{1}{z^2} \\
&+ (A + Bi - C + Di)\,\tfrac{1}{z^3} = 0;
\end{aligned}$$

in which z may be considered as the variable parameter; hence the equation of the caustic may be obtained by equating to zero the discriminant of the above function of z; but the discriminant of a sextic function has not yet been calculated. The equation would be of the order 20, and it appears from the result previously obtained for parallel rays, that the equation must be of the order 12 at the least; it is, I think, probable that there is not any reduction of order in the general case. It is however practicable, as will presently be seen, to obtain the tangential equation of the caustic by refraction, and the curve is thus shown to be only of the class 6.

XXVII.

Suppose that rays proceeding from a point are refracted at a circle, and let it be required to find the equation of the secondary caustic: take the centre of the circle as origin, let c be the radius, ξ, η the coordinates of the radiant point, α, β the coordinates of a point upon the circle, μ the index of refraction; the secondary caustic will be the envelope of the circle,

$$\mu\{(x-\alpha)^2+(y-\beta)^2\}-\{(\xi-\alpha)^2+(\eta-\beta)^2\}=0,$$

where α, β are variable parameters connected by the equation $\alpha^2+\beta^2-c^2=0$; the equation of the circle may be written in the form

$$\mu^2(x^2+y^2+c^2)-(\xi^2+\eta^2+c^2)-2(\mu^2x-\xi)\alpha-2(\mu^2y-\eta)\beta=0.$$

But in general the envelope of $A\alpha+B\beta+C=0$, where α, β are connected by the equation $\alpha^2+\beta^2-c^2=0$, is $c^2(A^2+B^2)-C^2=0$, and hence in the present case the equation of the envelope is

$$\{\mu^2(x^2+y^2+c^2)-(\xi^2+\eta^2+c^2)\}^2=4c^2\{(\mu^2x-\xi)^2+(\mu^2y-\eta)^2\},$$

which may also be written

$$\{\mu^2(x^2+y^2-c^2)-(\xi^2+\eta^2-c^2)\}^2=4c^2\mu^2\{(x-\xi)^2+(y-\eta)^2\}.$$

If the axis of x be taken through the radiant point, then $\eta=0$, and writing also $\xi=a$, the equation becomes

$$\{\mu^2(x^2+y^2-c^2)-a^2+c^2\}^2=4c^2\mu^2\{(x-a)^2+y^2\};$$

or taking the square root of each side,

$$\{\mu^2(x^2+y^2-c^2)-a^2+c^2\}=2c\mu\sqrt{(x-a)^2+y^2};$$

whence multiplying by $\dfrac{1}{\mu^2}$ and adding on each side $c^2\left(\mu-\dfrac{1}{\mu}+\dfrac{a}{\mu}\right)(x-a)^2+y^2$, we have

or

$$\sqrt{\left(x-\tfrac{a}{\mu}\right)^2+y^2}=\sqrt{(x-a)^2+y^2}+c\left(\mu-\tfrac{1}{\mu}\right),$$

which shows that the secondary caustic is the Oval of Descartes, or as it will be convenient to call it, the Cartesian.

It is proper to remark, that the Cartesian consists in general of two ovals, one of which is the orthogonal trajectory of the refracted rays, the other the orthogonal trajectory of the false refracted rays. In the case of reflexion, the secondary caustic is a Cartesian having a double point; this may be either a conjugate point, or a real double point arising from the union and intersection of the two ovals; the same secondary caustic may arise also from refraction, as will be presently shown.

XXVIII.

Reverting to the original form of the equation of the secondary caustic, multiplying by $\dfrac{1}{\mu^2}\left(1-\dfrac{c^2}{a^2}\right)$ and adding on each side $\dfrac{c^2}{\mu^2}\left(1-\dfrac{c^2}{a^2}\right)+\dfrac{c^2}{a^2}\{(x-a)^2+y^2\}$, the equation becomes

$$\left\{\left(x-\tfrac{c^2}{a}\right)^2+y^2\right\}=\left[\tfrac{c}{a}\sqrt{(x-a)^2+y^2}+\tfrac{a}{\mu}\left(1-\tfrac{c^2}{a^2}\right)\right]^2,$$

or extracting the square root,

$$\sqrt{\left(x-\tfrac{c^2}{a}\right)^2+y^2}=\sqrt{(x-a)^2+y^2}+\tfrac{a}{\mu}\left(1-\tfrac{c^2}{a^2}\right).$$

Combining this with the former result, we see that the equation may be expressed indifferently in any one of the four forms,

$$\sqrt{\left(x-\tfrac{a}{\mu}\right)^2+y^2}=\tfrac{1}{\mu}\sqrt{(x-a)^2+y^2}+c\left(\mu-\tfrac{1}{\mu}\right),$$
$$\sqrt{\left(x-\tfrac{c^2}{a}\right)^2+y^2}=\tfrac{c}{a}\sqrt{(x-a)^2+y^2}+\tfrac{a}{\mu}\left(1-\tfrac{c^2}{a^2}\right),$$
$$\sqrt{\left(x-\tfrac{c^2}{a}\right)^2+y^2}=\tfrac{c\mu}{a}\sqrt{\left(x-\tfrac{a}{\mu}\right)^2+y^2}+\tfrac{a}{\mu}-\tfrac{c^2\mu}{a},$$

$$c\left(\mu-\tfrac{1}{\mu}\right)\sqrt{\left(x-\tfrac{c^2}{a}\right)^2+y^2}+\left(-a+\tfrac{c^2}{a}\right)\sqrt{\left(x-\tfrac{a}{\mu}\right)^2+y^2}+\left(\tfrac{a}{\mu}-\tfrac{c^2\mu}{a}\right)\sqrt{(x-a)^2+y^2}=0.$$

It follows, that if we write successively

$a'=a,$	$c'=c,$	$\mu'=\mu$	(1)
$a'=\frac{c^2}{a},$	$c'=\frac{c}{\mu},$	$\mu'=\frac{c}{a}$	(α)
$a'=\frac{a}{\mu^2},$	$c'=\frac{c}{\mu},$	$\mu'=\frac{1}{\mu}$	(β)
$a'=a,$	$c'=\frac{c}{\mu},$	$\mu'=\frac{a}{c\mu}$	(γ)
$a'=\frac{c^2}{a},$	$c'=c,$	$\mu'=\frac{c\mu}{a}$	(δ)
$a'=\frac{c^2}{a},$	$c'=\frac{c}{\mu},$	$\mu'=\frac{a}{c\mu}$	(ε),

or what is the same thing,

$$\begin{array}{llll}
a = a', & c = c', & \mu = \mu' & (1) \\
a = \frac{c'^2}{a'}, & c = \frac{c'}{\mu'}, & \mu = \frac{a'}{c'} & (\alpha) \\
a = \frac{a'}{\mu'^2}, & c = \frac{c'}{\mu'}, & \mu = \frac{1}{\mu'} & (\beta) \\
a = a', & c = \frac{c'}{\mu'}, & \mu = \frac{a'}{c'\mu'} & (\gamma) \\
a = \frac{c'^2}{a'}, & c = c', & \mu = \frac{c'\mu'}{a'} & (\delta) \\
a = \frac{c'^2}{a'}, & c = \frac{c'}{\mu'}, & \mu = \frac{c'}{a'} & (\varepsilon).
\end{array}$$

or what is again the same thing,

$$\begin{array}{llll}
a' = a, & \frac{c'^2}{a'} = \frac{c^2}{a}, & \frac{a'}{\mu'^2} = \frac{a}{\mu^2} & (1) \\
a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = a, & \frac{a'}{\mu'^2} = \frac{c^2}{\mu^2} & (\alpha) \\
a' = \frac{a}{\mu^2}, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = a & (\beta) \\
a' = a, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = \frac{c^2}{a} & (\gamma) \\
a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = a, & \frac{a'}{\mu'^2} = \frac{a}{\mu^2} & (\delta) \\
a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = \frac{c^2}{a} & (\varepsilon),
\end{array}$$

we have in each case identically the same secondary caustic, and therefore also identically the same caustic; in other words, the same caustic is produced by six different systems of a radiant point and refracting circle. It is proper to remark that if we represent the six systems of equations by $(a', c', \mu') = (a, c, \mu)$, $(a', c', \mu') = \alpha(a, c, \mu)$, &c., then $\alpha, \beta, \gamma, \delta, \varepsilon$ will be functional symbols satisfying the conditions

$$\begin{aligned}
&1 = \alpha\beta = \beta\alpha = \gamma^2 = \delta^2 = \varepsilon^2, \\
&\alpha = \beta^2 = \delta\gamma = \varepsilon\delta = \gamma\varepsilon, \\
&\beta = \alpha^2 = \gamma\delta = \delta\varepsilon = \varepsilon\gamma, \\
&\gamma = \delta\alpha = \alpha\varepsilon = \varepsilon\beta = \beta\delta, \\
&\delta = \varepsilon\alpha = \alpha\gamma = \gamma\beta = \beta\varepsilon, \\
&\varepsilon = \gamma\alpha = \alpha\delta = \delta\beta = \beta\gamma.
\end{aligned}$$

XXIX.

The preceding formulae, which were first given by me in the *Philosophical Magazine*, December 1853, [124] include as particular cases a preceding theorem with respect to the caustic by refraction of parallel rays, and also two theorems of St Laurent, *Gergonne*, t. XVIII., [1827, pp. 1–19] viz. if we suppose first that $a = c$, i.e. that the radiant point is in the circumference of the refracting circle, then the system (α) shows that the same caustic would be obtained by writing $c, \frac{c}{\mu}, 1$ (or what is the same thing $\frac{1}{\mu} = 1$) in the place of c, c, μ, and we have

Theorem. The caustic by refraction for a circle when the radiant point is in the circumference is also the caustic by reflexion for the same radiant point, and for a reflecting circle concentric with the refracting circle, but having its radius equal to the quotient of the radius of the refracting circle by the index of refraction.

Next, if we write $a = c\mu$, then the refracted rays all of them pass through a point which is a double point of the secondary caustic, the entire curve being in this case the orthogonal trajectory, not of the refracted rays, but of the false refracted rays; the formula (δ) shows that the same caustic is obtained by writing $\frac{c}{\mu}, c, 1$ (or what is the same thing $\mu = 1$) in the place of $a, c, \mu\left(= \frac{a}{c}\right)$, and we have

Theorem. The caustic by refraction for a circle when the distance of the radiant point from the centre is to the radius of the circle in the ratio of the index of refraction to unity, is also the caustic by reflexion for the same circle considered as a reflecting circle, and for a radiant point the image of the former radiant point.

XXX.

The curve is most easily traced by means of the preceding construction thus; if we take the radiant point outside the refracting circle, and consider μ as varying from a small to a large value (positive or negative values of μ give the same curve), we see that when μ is small the curve consists of two ovals, one of them within and the other without the refracting circle (see fig. 14). As μ increases the exterior oval continually increases, but undergoes modifications in its form; the interior oval in the first instance diminishes until we arrive at a curve, in which the interior oval is reduced to a conjugate point (see fig. 15); then as μ continues to increase the interior oval reappears (see fig. 16), and at last connects itself with the exterior oval, so as to form a curve with a double point (see fig. 17); and as μ increases still further the curve again breaks up into an exterior and an interior oval (see fig. 18); and thenceforward as μ goes on increasing consists always of two ovals; the shape of the exterior oval is best perceived from the figures. An examination of the figures will also show how the same curves may originate from a different refracting circle and radiant point.

Fig. 14.

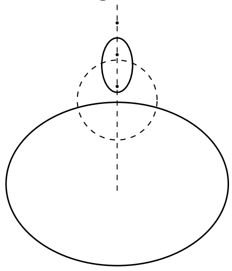


Fig. 17.

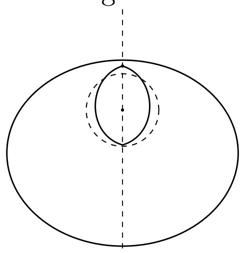


Fig. 15.

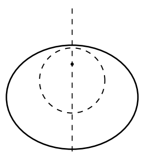


Fig. 16.

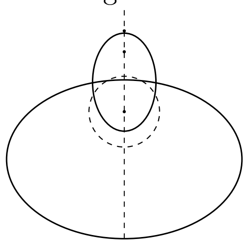
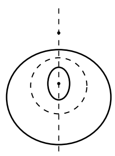


Fig. 18.



XXXI.

The theorem, "If a variable circle have its centre upon a circle S, and its radius proportional to the tangential distance of the centre from a circle C, the envelope is a Cartesian," is at once deducible from the theorem

"If a variable circle have its centre upon a circle S and its radius proportional to the distance of the centre from a point C, the locus is a Cartesian,"

which last theorem was in effect given in discussing the theory of the secondary caustic. In fact, the locus of a point P such that its tangential distances from the circles C, C' are in a constant ratio, is a circle S. Conversely, if there be a circle C, and the locus of P be a circle S, then the circle C' may be found such that the tangential distances of P from the two circles are in a constant ratio, and the circle C' may be taken to be a point, i.e. if there be a circle C and the locus of P be a circle S, then a point C' may be found such that the tangential distance of P from the circle C is in a constant ratio to the distance from the point C'.

Hence treating P as the centre of the variable circle, it is clear that the variable circle is determined in the two cases by equivalent constructions, and the envelope is therefore the same in both cases.

XXXII.

The equation of the secondary caustic developed and reduced is

$$\mu^4(x^2+y^2)^2 - 2\mu^2(a^2+(\mu^2+1)c^2)(x^2+y^2) + 8c^2\mu^2a^2x + a^4 - 2a^2c^2(\mu^2+1) + (\mu^2-1)^2c^4 = 0,$$

or, what is the same thing,

$$\{\mu^2(x^2+y^2) - (a^2+(\mu^2+1)c^2)\}^2 + 8c^2\mu^2a^2x - 4c^2(c^2\mu^2+(\mu^2+1)a^2) = 0,$$

which may also be written

$$\left(x^2+y^2-\left(\tfrac{a^2}{\mu^2}+(1+\tfrac{1}{\mu^2})c^2\right)\right)^2 + \tfrac{8}{\mu^2}c^2a^2x - \tfrac{4c^2}{\mu^4}\left(c^2+(1+\tfrac{1}{\mu^2})a^2\right) = 0,$$

which is of the form

$$(x^2+y^2-\alpha)^2 + 16A(x-m) = 0,$$

and the values of the coefficients are

$$\alpha = \tfrac{a^2}{\mu^2} + (1+\tfrac{1}{\mu^2})c^2,$$

$$A = \tfrac{c^2a}{2\mu^2},$$

$$m = \tfrac{1}{2a}\left(c^2+(1+\tfrac{1}{\mu^2})a^2\right).$$

The equation just obtained should, I think, be taken as the standard form of the equation of the Cartesian, and the form of the equation shows that the Cartesian may be defined as the locus of a point, such that the fourth power of its tangential distance from a given circle is in a constant ratio to its distance from a given line.

XXXIII.

The Cartesian is a curve of the fourth order, symmetrical about a certain line which it intersects in four arbitrary points, and these points determine the curve. Taking the line in question (which may be called the axis) as the axis of x, and a line at right angles to it as the axis of y, let a, b, c, d be the values of x corresponding to the points of intersection with the axis, then the equation of the curve is

$$y^4 + y^2\big[2x^2 - (a+b+c+d)x - \tfrac{1}{2}(a^2+b^2+c^2+d^2-2ab-2ac-2ad-2bc-2bd-2cd)\big] + (x-a)(x-b)(x-c)(x-d) = 0.$$

It is easy to see that the form of the equation is not altered by writing $x+\theta$ for x, and $a+\theta, b+\theta, c+\theta, d+\theta$ for a, b, c, d; we may therefore without loss of generality put $a+b+c+d=0$, and the equation of the curve then becomes

$$y^4 + y^2(2x^2+ab+ac+ad+bc+bd+cd) + (x-a)(x-b)(x-c)(x-d) = 0,$$

where

$$a+b+c+d=0;$$

the curve is in this case said to be referred to the centre as origin.

The last-mentioned equation may be written

$$(x^2+y^2+\tfrac{1}{2}(ab+ac+ad+bc+bd+cd))(x^2+y^2) - (abc+abd+acd+bcd)x + abcd = 0,$$

$$\begin{aligned}&\{x^2+y^2+\tfrac{1}{2}(ab+ac+ad+bc+bd+cd)\}^2\\ &\quad -(abc+abd+acd+bcd)\,x\\ &\quad +\tfrac{1}{4}\big[\,a^2b^2+a^2c^2+a^2d^2+b^2c^2+b^2d^2+c^2d^2\\ &\qquad\quad +2a^2bc+2a^2bd+2a^2cd+2b^2ac+2b^2ad+2b^2cd\\ &\qquad\quad +2c^2ab+2c^2ad+2c^2bd+2d^2ab+2d^2ac+2d^2bc\\ &\qquad\quad +2abcd\,\big]=0,\end{aligned}$$

or observing that

$$\begin{aligned}&a^2bc+a^2bd+a^2cd+b^2ac+b^2ad+b^2cd\\ &\quad +c^2ab+c^2ad+c^2bd+d^2ab+d^2ac+d^2bc\\ &=abc(a+b+c)+abd(a+b+d)+acd(a+c+d)+bcd(b+c+d)\\ &=-4abcd,\end{aligned}$$